

Data Collection and Exploratory Data Analysis

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- 6 Many Variables
- 7 Time and Space

Data Sources and Types

Where data come from

Source	Examples
Surveys, administrative records	DHS, MICS, census extracts, hospital or school records
Sensors and logs	Weather stations, phones, server logs
Web and APIs	Scraped pages, social media, open data
Remote sensing	Satellite imagery, NDVI

How data are structured

- **Structured:** fixed rows and columns with a schema, such as survey microdata, spreadsheets, and relational tables
- **Semi-structured:** self-describing hierarchy without a fixed table layout, such as JSON from APIs, XML, and server logs; it must be parsed into columns before analysis
- **Unstructured:** no predefined layout, such as free text, images, audio, and video; useful numbers must be extracted first

Variable types decide the analysis

Type	Example	Summaries	Plots
Nominal	gender, religion	counts, mode	bar chart
Ordinal	socio-economic status, class	counts, median	ordered bars
Discrete	number of children	counts, mean	bar chart
Continuous	price, age	mean, median, SD	histogram, boxplot

- Survey microdata often store categories as numeric codes (1 = male, 2 = female, 99 = don't know); recognise them before summarising
- The type decides which summary is meaningful: gender could be stored as 1, 2, yet an 'average gender' of 1.5 means nothing

Why Exploratory Data Analysis?

What is EDA?

- Exploratory data analysis (EDA) is the systematic examination of a dataset through summary statistics and graphics before any formal modelling, in the spirit of John Tukey's "detective work".
- It asks what each variable looks like, how variables relate to each other, which values are unusual, and whether there is structure such as groups or trends.
- Its output is decisions: which questions the data can answer, which problems must be fixed first, and which kinds of model, if any, are worth trying.
- EDA is not confirmatory: patterns found this way are hypotheses, to be tested on data that were not used to find them.

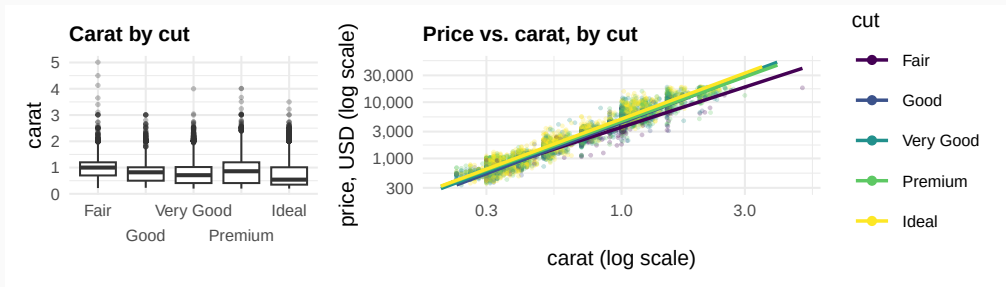
Why look at the data? An example

Diamonds: 53,940 stones, price in US dollars. Mean price by cut, from worst to best:

Cut	n	Mean price	Mean carat
Fair	1,610	\$4,359	1.05
Good	4,906	\$3,929	0.85
Very Good	12,082	\$3,982	0.81
Premium	13,791	\$4,584	0.89
Ideal	21,551	\$3,458	0.70

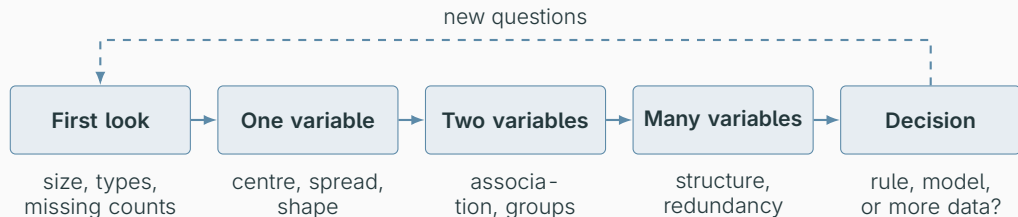
- The best cut (Ideal) has the lowest mean price, and the worst cut (Fair) is dearer than Good, Very Good, and Ideal
- Taken at face value, a better cut lowers the price, and a model using cut alone would learn exactly that
- The last column hints at the reason: Ideal stones are the smallest on average (0.70 carat) and Fair the largest (1.05)

Why look at the data? The plots



- Left: Ideal stones are mostly small (median 0.54 carat) and Fair stones large (median 1.00 carat)
- Right: within every cut, price rises steeply with carat; at exactly 1 carat, Fair costs \$4,130 on average ($n = 166$) and Ideal \$5,547 ($n = 208$)

The EDA workflow



- Each stage raises questions for the next, so EDA is iterative rather than a fixed pipeline
- The decision at the end is not always “fit a model”: it may be to collect more data, apply a simple rule rather than a complex ML algorithm, or report the summary

First look at a dataset

- Size and types: number of rows and columns and the type of each variable
- Missing values: count them per variable; take necessary actions in case of any missing values
- Duplicate rows and impossible values, such as negative ages or zero dimensions

One Variable at a Time

Location: mean, median, trimmed mean, mode

Observations $x_1, \dots, x_n$ of one numeric variable.

- **Mean** $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$: uses every value, so extreme values pull it
- **Median**: the middle value of the sorted data, with half of the observations on each side; extreme values do not move it, so it is the better measure of centre for skewed data such as prices, incomes, or waiting times
- **Trimmed mean**: the mean after discarding a fixed share (say 10%) of observations from each tail
- **Mode**: the most frequent value

Spread and shape

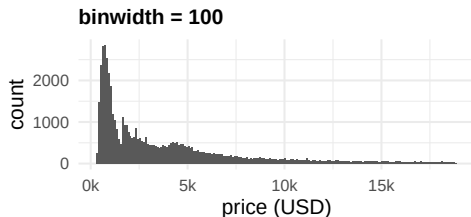
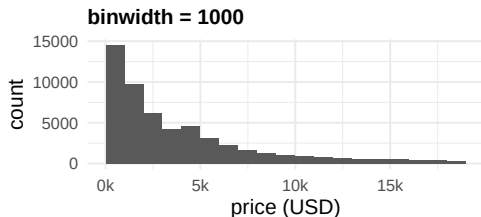
- **Standard deviation:** $s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$: typical distance from the mean; sensitive to extreme values
- **Interquartile range:** third quartile minus first quartile, $IQR = Q_3 - Q_1$: width of the middle 50% of the data; robust to extreme values
- **Skewness:** $g_1 = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s}\right)^3$: positive for a long right tail, negative for a long left tail, near zero for a symmetric distribution
- **Kurtosis:** $g_2 = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s}\right)^4 - 3$: positive when the tails are heavier than those of a normal distribution

Summary statistics in practice: diamonds

	price (USD)	carat
Mean	3,933	0.80
10% trimmed	3,159	0.73
Median	2,401	0.70
Std. dev.	3,989	0.47
IQR	4,374	0.64
MAD	1,670	0.32
Skewness	1.62	1.12
Ex. kurtosis	2.18	1.26

- The mean price is 64% above the median: a few expensive stones pull the mean up, while the median describes a typical stone
- Skewness of 1.62 confirms a long right tail; mean $>$ median and positive skewness go together

Histograms and density estimates



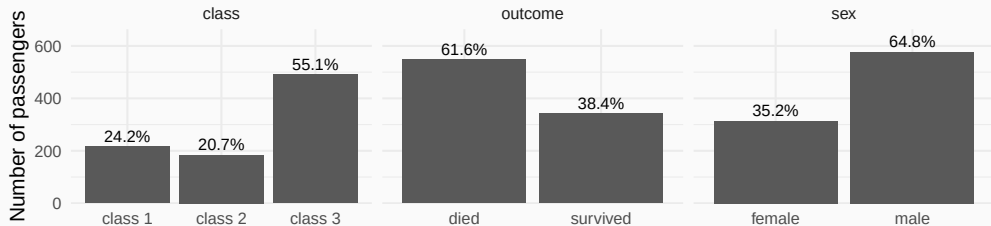
- **Histogram:** counts in equal-width bins; try several widths (wide bins hide structure, narrow bins add noise); read peaks, gaps, and spikes
 - At width 100 the count falls to 463 in \$1,500–1,599 and jumps to 1,115 in the next bin, a jump that width 1000 hides
 - Use with hundreds of observations or more; for small samples use a dot or strip plot

Boxplot, violin plot, and ECDF



- **Boxplot:** box from Q_1 to Q_3 with the median line; whiskers reach the furthest points within $1.5 \times \text{IQR}$; points beyond are flagged (3,540 stones, 6.6%, all expensive)
 - Use to compare many groups side by side; it cannot show several peaks

Categorical variables: frequencies and bar charts



- A frequency table lists the count and share of each level, and the bar chart is its picture
- Titanic, 891 passengers: 61.6% died and 38.4% survived; 55.1% travelled third class; 64.8% were male
- Imbalance matters: predicting “died” for everyone is already 61.6% accurate, a baseline any model must beat
- Order nominal levels by frequency and ordinal ones naturally

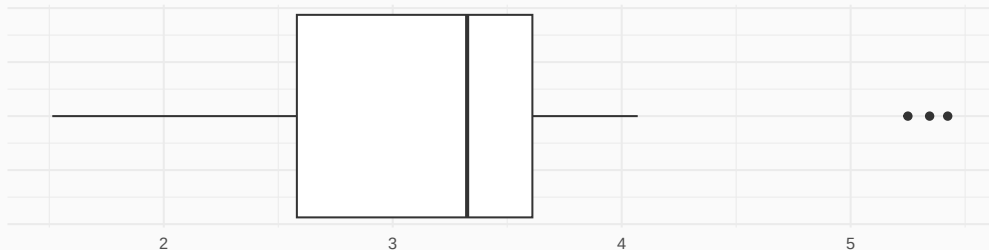
Data Quality

Missing values: count first, then decide

Dataset	Variable	Missing	Share
Titanic (891 rows)	age	177	19.9%
	deck	688	77.2%
	embarked	2	0.2%
California housing (20,640)	total_bedrooms	207	1.0%

- Count the missing values per variable: 2 missing values (embarked) call for a different response than 77% (deck)
- Ask why: Titanic age is missing for 27.7% of third-class but 6.0% of second-class passengers, so not at random; total_bedrooms is missing at 0.8–1.1% in each large ocean-proximity group, no visible pattern
- Missing values can hide as codes (99, -1, 0), common in survey microdata; check value counts
- If any are present, judge whether action is required and take the necessary steps (preprocessing lecture)

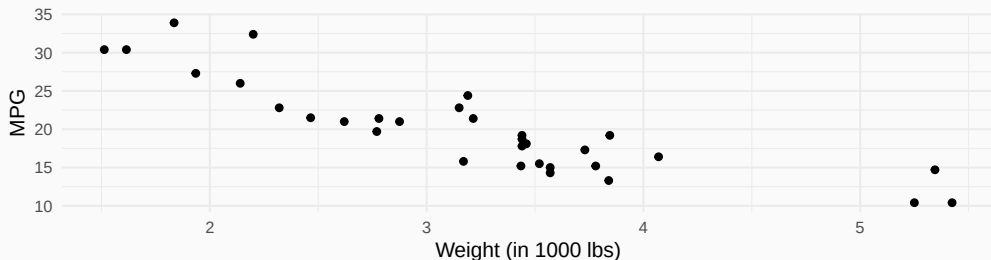
Detecting outliers



- **$1.5 \times \text{IQR}$ rule:** flag values below $Q_1 - 1.5 \text{ IQR}$ or above $Q_3 + 1.5 \text{ IQR}$
- **z -score:** $z_i = (x_i - \bar{x})/s$, flag $|z_i| > 3$; the mean and SD are pulled by the outliers themselves, which can mask them
- **Modified z -score:** $M_i = 0.6745 (x_i - \tilde{x})/\text{MAD}$, flag $|M_i| > 3.5$; robust, but on skewed data it flags the long tail

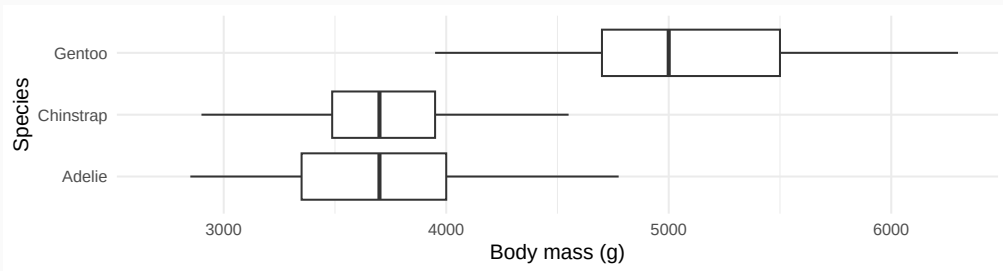
Two Variables

Two numeric variables: scatter plots



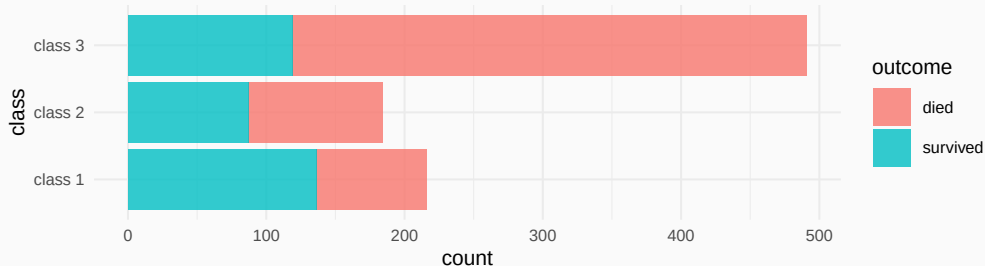
- A scatter plot puts each observation at (x, y)
- The overall shape of the plot can tell us whether there is any association between the two variables
- The correlation coefficient is a numeric measure that quantifies the linear relationship between the two variables

Numeric variable across groups



- Mean body mass: Adelie 3,701 g, Chinstrap 3,733 g, Gentoo 5,076 g; the first two are nearly equal, Gentoo is about 1.4 kg heavier
- Mean $\pm$ 1.96 SE intervals (about ± 73 , ± 91 , ± 89 g) overlap for Adelie and Chinstrap; ANOVA gives $F(2, 339) = 343.6$, $\eta^2 = 0.67$: species explains 67% of the variance

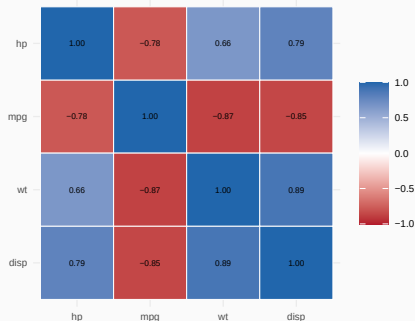
Two categorical variables



- Above is a stacked barplot
- It shows the proportion of survived passengers among different passenger classes in the Titanic

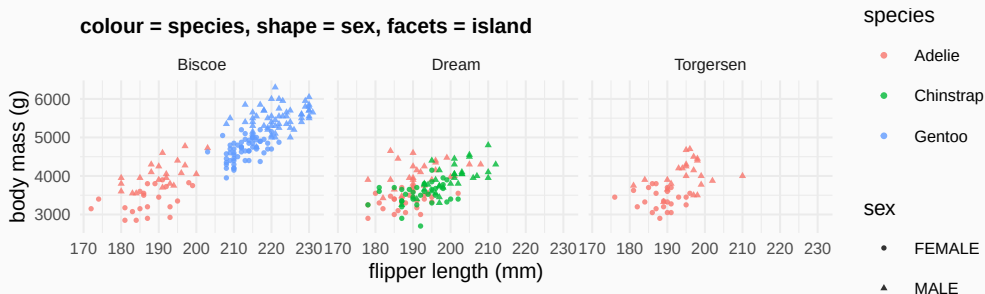
Many Variables

Correlation heatmap and multicollinearity



- Heatmap: a tile per pair, diverging colours centred at zero; clustering the order puts related variables in blocks
- mtcars: mpg vs. weight -0.87 , cylinders -0.85 , displacement -0.85 ; displacement vs. cylinders 0.90 , weight 0.89
- Multicollinearity: overlapping predictors; $VIF_j = 1/(1 - R_j^2)$, R_j^2 from regressing predictor j on the others; cyl 6.7, disp 10.4, hp 3.4, wt 4.8 (concern above about 5–10)
- Shows pairwise linear association only

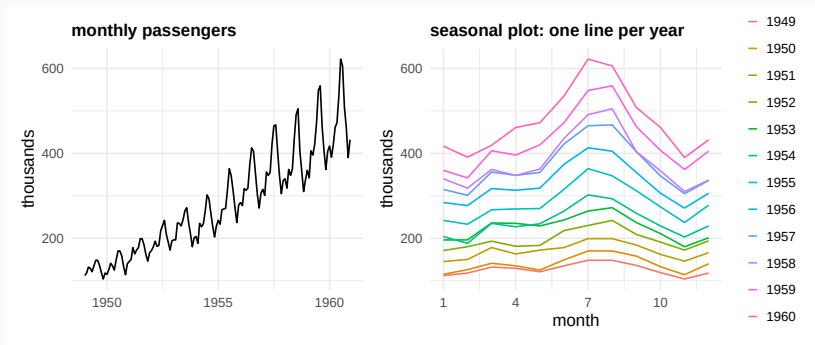
More dimensions: colour, shape, facets, parallel coordinates



- Extra variables can be encoded by colour (categorical), size or a colour gradient (numeric), shape, or faceting into small multiples with shared scales; keep to three or four variables per plot
- Facets by island reveal structure: Gentoo lives only on Biscoe and Chinstrap only on Dream, while Adelie is on all three, so island and species are entangled
- Parallel coordinates draw one line per observation across standardised variables; species form bands (Gentoo: shallow bills, long flippers, high mass), but many

Time and Space

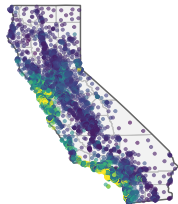
Time series: trend, seasonality, variance



- Plot against time first; look for trend, seasonality, changing variance, and breaks; the yearly mean rises from 126.7 to 476.2 thousand over 1949–1960
- Seasonal peak in July and August

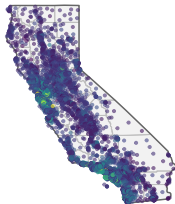
Spatial data: map before you model

Median house value (USD)



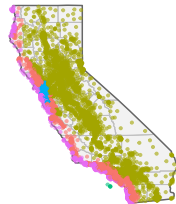
500k
400k
300k
200k
100k

Median income



15
10
5

Ocean proximity



<1H OCEAN
INLAND
ISLAND
NEAR BAY
NEAR OCEAN

- Map each variable (20,640 districts): values are high near the coast, low inland (median \$108,500 inland vs. \$214,850–\$233,800 coastal)
- Observations are correlated, nearby observations resemble each other
- Observations are not independent, so random train/test splits overstate accuracy
- Check coordinates and resolution: survey cluster locations (e.g. DHS) are deliberately displaced

Summary

EDA checklist and tools

- ① First look: data size, types, missing, duplicates
- ② Each variable: summaries, histogram or bar chart
- ③ Quality: outliers vs. artefacts, caps, codes
- ④ Pairs: plots, correlations, groups, confounders
- ⑤ Structure: time order, space, groups
- ⑥ Record findings and decisions in a script

Questions?
